ENGINEERING STATUS REPORT

Completed Features and Remaining Work

A detailed snapshot of the Rails backend and React frontend, organized as **21 completed features** and **8 not-completed features**.

21
Completed features

8
Not fully completed

72%
21 of 29 in agreed scope



21 completed 2 partial 6 not implemented

Scope and Counting

The original list contains 30 requirements. The India-versus-US comparison in a common currency (original #9) is excluded, leaving 29. Salary averages and expenditure rankings are currency-specific rather than converted into a common currency.

Priority	Completed	Not Fully Completed	Assessment
P0: MVP	17	0	Implemented within the adjusted reporting scope.
P1: High product value	4	2	Role comparisons, review reports and salary data-quality checks are available.
P2: Future enhancements	0	6	Planning, historical comparisons and export remain open.

> **Implementation is not deployment.** The latest reporting enhancements are implemented and tested locally but have not been committed, pushed or deployed. The most recent repository commit is `88c4571`, for single-HR authentication.

Verification Snapshot

358 Rails specs and **43 frontend tests** pass. The React production build, Ruby lint and Brakeman scan passed. Desktop browser checks used the real local database containing **10,032 employees**. Manual salary checks preserve employee data and HR review history.

Completion reflects implemented behavior, not production performance certification or conclusions about salary fairness. No credentials or employee-level salary records are included in this report.

Completed: Salary Overview and Analysis

Annual base salaries and monetary statistics remain separated by currency. Original requirement numbers are included for traceability.

No.	Completed Feature	Delivered Behavior and Scope
1	**Annual base salary totals** Original #1 | P0	Aggregates current annual base salaries per currency. Available in Overview and Compensation; country, department and status filters apply.
2	**Monthly payroll estimates** Original #2 | P0	Annual base-salary total divided by 12, calculated before rounding. Excludes taxes, benefits, bonuses, prorating and payroll scheduling.
3	**Total employee count** Original #3 | P0	Reports total headcount for the selected scope. Defaults include active and inactive staff; the active-status filter narrows the population.
4	**Average salary per currency** Original #4 | P0 | Adjusted scope	Reports annual arithmetic averages within each currency. This replaces the original normalized-average requirement; exchange-rate normalization is not implemented.
5	**Median salary per currency** Original #5 | P0	Supports odd and even employee counts. Even-count medians average the two middle salaries with decimal arithmetic.
6	**Highest and lowest salaries** Original #6 | P0	Returns minimum and maximum current annual salary per currency, country and department. Overview and Compensation display the currency-level ranges.
7	**Country expenditure rankings** Original #7 | P0	Ranks countries by annual total within the selected currency. Sorting can switch to annual average or headcount; there is no cross-currency monetary ranking.
8	**Department expenditure rankings** Original #8 | P0	Ranks departments within each currency, showing headcount, annual total and annual average. The highest-spending department appears first when sorted by total.
9	**Consolidated country headcounts** Original #10 | P0	Combines employee counts across currencies for each country and displays workforce share. Consolidated counts reconcile with overall headcount.
10	**Department salary bands** Original #11 | P0	Four fixed annual salary bands per currency, with department counts and currency totals. Lower bounds are inclusive, upper bounds exclusive; the final band is open-ended.
11	**Ranked department averages** Original #12 | P0	Choosing Average salary in the department ranking control identifies the highest average within the selected currency. Ties are resolved by department name.

Completed numbering continues on the next page

Completed: Employee Workflows and Review

No.	Completed Feature	Delivered Behavior and Scope
12	**Employee search** Original #13 | P0	Searches by employee name or employee ID. Results use the paginated employee directory and combine with the existing filters.
13	**Current salary details** Original #14 | P0	Shows employee identity, department, role, country, status, current annual base salary, currency and effective date through the salary drawer and API.
14	**Salary updates** Original #15 | P0	Salary changes require review and explicit confirmation. Backend validation and transactional updates preserve old/new values and salary history.
15	**Combined employee filters** Original #16 | P0	Country, department and status filters combine. The same filters apply to compensation reports and salary-analysis comparison populations.
16	**Salary sorting** Original #17 | P0	Supports ascending and descending numeric salary sorting. Employee ID breaks salary ties for predictable pagination.
17	**10,000-employee handling** Original #18 | P0	Server-side pagination, bounded pages and database indexes are implemented. Tests exercise 10,000 employees; live local checks use 10,032. Production latency and concurrent load are not certified.
18	**Role-based salary comparisons** Original #19 | P1	Groups employees by role (designation), country and currency. Shows group size, average, median, lowest/highest salary and salary gap. Group results are server-paginated.
19	**High-salary outlier reports** Original #20 | P1	Flags unusually high salaries in same-role/country/currency cohorts of at least five employees. Shows peer average, review threshold and excess amount; HR can open the existing salary drawer.
20	**Below-department-average lists** Original #21 | P1	Lists employees whose salary is strictly below the unrounded department average within the same currency. Shows group size and shortfall; baseline calculations precede pagination.
21	**Salary data-quality anomalies** Original #23 | P1	Eight manual checks with stored evidence, filtered review queue, HR decisions and immutable notes history. Rechecks preserve unchanged decisions; corrected findings clear without deleting history. Details and routes: pages 6 and 9.

> **Outlier rule:** salary must exceed Q3 + 1.5 x (Q3 - Q1), using discrete quartiles. When Q1 equals Q3, the threshold is 1.5 x Q3. The five-person cohort includes the employee under review. A flag is an investigation indicator, not evidence of unfair pay.

Additional supporting capabilities, not counted separately: single-HR authentication, new employee creation, salary-change history and the public API status page with light/black themes.

Not Fully Completed: Remaining Work

This list includes two partially implemented areas and six features without a dedicated implementation. Country/role fairness is unchanged and deferred by request.

No.	Remaining Feature	Status and Missing Work
1	**Country/role salary-fairness analysis** Original #22 | P1	**Partial.** Basic comparable role/country/currency groups exist. A dedicated fairness analysis with contextual evidence and explanations for differences is not implemented. Experience, seniority, hours and performance are not captured.
2	**Department salary dispersion** Original #24 | P1	**Partial.** Department minimum/maximum salaries and salary bands exist. Standard deviation, variance, IQR and a ranked comparison of department spread are not implemented.
3	**Organization-wide raise simulation** Original #25 | P2	**Not implemented.** Needs a read-only scenario for a percentage increase, reporting incremental annual/monthly cost per currency without modifying employee salaries.
4	**Department raise simulation** Original #26 | P2	**Not implemented.** Needs department-specific raise scenarios with selected scope, percentage input and projected costs, separated by currency.
5	**Year-over-year salary expenditure** Original #27 | P2	**Not implemented.** Salary-change history exists, but there is no year-end expenditure reconstruction, annual snapshot comparison or treatment of changing employee populations.
6	**Salary review schedule** Original #28 | P2	**Not implemented.** Needs review dates or review policy, overdue/upcoming lists and workflow status. A salary effective date is not equivalent to a scheduled review date.
7	**Upcoming increment budgets** Original #29 | P2	**Not implemented.** Needs planned increases, intended effective dates and scoped forecasts, with currency-specific budget totals.
8	**CSV or Excel report export** Original #30 | P2	**Not implemented.** Needs an authenticated export flow with selected filters, currency labels, protected salary data and spreadsheet formula-injection safeguards.

> **Separate exclusion:** India-versus-US comparison in a common currency (original #9) is outside this MVP and is not one of the eight remaining items. Exchange-rate conversion and normalized cross-currency averages are not provided.

What Has Been Verified

Check	Result	Coverage / Limitation
Rails RSpec	**358 examples, 0 failures**	Models, services, requests, authentication, status/health, seeds, deployment and 10,000-record anomaly checks.
Frontend Playwright	**43 tests passed**	5 API-client tests and 38 desktop browser tests, including 5 backend theme tests. Portal workflow tests mock API responses.
React production build	**Passed**	TypeScript checking and Vite production build.
Ruby lint	**80 files, no offenses**	Repository RuboCop configuration.
Brakeman	**0 security warnings**	Static scan; not a substitute for deployment or penetration testing.
Real local browser checks	**Passed**	Real Rails data, 10,032 employees; desktop widths 1280, 1440 and 1920; no browser errors observed.
Local report responses	**15 / 28 / 16 ms**	Observed role / outlier / below-average responses on one local run, not production benchmarks or load-test percentiles.
Real salary check	**10,032 checked**	339 ms for one local UI/API scan; zero findings. Not a production latency guarantee; employee data was unchanged.

Application Route Reference

Pages 7-9 provide the complete application-route reference, including methods, uses, public versus HR access, query parameters and request examples. It was checked against `Backend/config/routes.rb` and `rails routes --expanded`.

Public Status and Health

The backend root serves an HTML status page with native light/black appearance, a saved theme preference, application liveness, UTC check time and a React portal link. JSON health remains available and Render continues to use `/up`. These checks do not verify database connectivity or expose employee records.

The public status page no longer displays API endpoints or version/base-path metadata. Route documentation remains in this report and the repository contract. Hiding documentation does not replace authentication; HR API access remains protected.

Deployment State and Next Step

The latest reporting, status/theme and data-quality code is local and has not been committed or pushed. Deploy both services after review. Data quality requires the new review-table migration; Render's migration-enabled build script applies it. No employee seed is needed.

Production rendering, deployment health, concurrency and latency must be checked after release. Monetary comparisons use current stored annual salaries, not a historical payroll ledger. No currency conversion is performed.

Salary Data-Quality Anomaly Detection

This feature asks: **"Does this salary record look like a possible data mistake?"** It flags suspicious-but-valid records for HR review, not conclusions about overpay or underpay.

Possible Anomaly	Example	Important Caveat
Extra or missing zero	INR 1,200,000 entered as INR 12,000,000.	The amount is valid and nonnegative, so validation alone cannot identify the mistake.
Annual/monthly confusion	INR 100,000 monthly pay stored as the annual base salary instead of INR 1,200,000.	A twelvefold difference suggests a check, not proof of an error.
Incorrect currency	A salary intended as INR 1,200,000 is entered as USD 1,200,000.	Both currency codes are supported. Country alone cannot prove which currency is correct.
Unexpected zero salary	An active paid employee has an annual base salary of zero.	Zero is allowed today. Unpaid roles or unusual arrangements may be legitimate.
Abrupt salary change	A recorded change increases a salary tenfold or reduces it by 90%.	Currency changes, promotions and corrections require context; do not compare raw amounts across currencies.
Date or history inconsistency	A current salary conflicts with the latest recorded salary change.	Future effective dates can be legitimate. Initial salaries have no change entry, and audit timestamps must be interpreted correctly.

Validation vs Outlier vs Data-Quality Detection

Validation rejects invalid inputs. **Outlier reporting** highlights unusual peer-group pay. **Data-quality checks** flag possible amount, unit, currency, date or history mistakes. None automatically corrects salaries.

Delivered Checks and Limitations

Eight checks cover active zero pay; tenfold peer ratios; twelvefold changes (2% tolerance); fivefold increases or 80% drops; unchanged amounts on currency switches; dates over 365 days ahead; history mismatches; and blank details. Peer groups need five employees sharing country, role and currency. No conversion or fairness inference is performed.

Implemented HR Review Workflow

1. Choose Data quality > Run checks. Fixed rules inspect all employees in 500-record batches.
2. Show employee, reason code, observed value, reference value, severity and detection date.
3. Record Pending, Investigating, Confirmed error or Legitimate, with an immutable author/time audit trail. The last two require notes.
4. Correct through the confirmed salary-update flow, then rescan to clear flags without deleting history. Unchanged reruns preserve decisions.

> **Never auto-correct a flagged salary.** A flag is not proof of an error or unfair pay. HR confirms corrections through the existing salary workflow, then reruns checks.

Application Route Reference

Paths are relative to the Rails backend. Public routes do not require a session; HR routes require an active superadmin bearer token. JSON root health is a representation of the root route, not a separate controller action.

Method	Route	Use	Access
GET	/	Public application status, UTC check time, light/black theme and React portal link. No API route list or version/base-path metadata.	Public
GET	/?format=json	Machine-readable application liveness: status and timestamp. Also available with an application/json Accept header.	Public
GET	/up	Rails boot/liveness check used by Render. Supports HTML and JSON negotiation; does not test PostgreSQL.	Public
POST	/api/v1/session	Authenticate the HR user. Returns public user details, an opaque session token and an eight-hour expiry.	Public*
GET	/api/v1/session	Verify the bearer session and return the signed-in user's public details and expiry.	HR
DELETE	/api/v1/session	Sign out and revoke the current server-side session. Returns 204 with no response body.	HR
GET	/api/v1/employees	Search, filter, paginate and sort the employee directory by current salary.	HR
POST	/api/v1/employees	Create an employee with validated identity, department, country, role and initial salary details.	HR
GET	/api/v1/employees/:id	Read one employee's details and current salary.	HR
GET	/api/v1/employees/:employee_id/salary	Read the employee's current base salary, currency and effective date.	HR
PATCH PUT	/api/v1/employees/:employee_id/salary	Apply a validated, explicitly confirmed salary change and preserve the salary-change audit record.	HR
GET	/api/v1/employees/:employee_id/salary_history	Read that employee's salary changes, old/new values, effective dates and change timestamps.	HR
GET	/api/v1/compensation_insights	Currency totals, averages, medians, monthly estimates, min/max salaries, group salary bands and country headcounts.	HR
GET	/api/v1/salary_analysis	Paginated role comparisons, high-salary review indicators or below-department-average employees, selected by report parameter.	HR

*** Public login means no existing token is needed, not anonymous access to salary data.** Login attempts are rate-limited; valid credentials are still required.

> **Employee identifier:** both `:id` and the nested-route `:employee_id` mean the numeric database primary key returned as `id`, such as `42`. They do not mean the employee's business ID such as `SEED00042`.

Data-quality routes continue on page 9

Using the Routes

Authenticated requests: include `Authorization: Bearer <session-token>`. Send `Content-Type: application/json` for JSON request bodies. Never include real passwords or tokens in reports or source control.

Endpoint / Parameters	Use and Example
`POST /api/v1/session`	Body: `{"session":{"email":"<HR email>","password":"<HR password>"}}`. Use the returned token for HR API requests. A repeated seed does not automatically replace an existing user's password.
`GET /api/v1/employees` `q, country, department, status, page, per_page, sort, direction`	`/api/v1/employees?q=Alice&country=India&status=active&sort=base_salary&direction=desc&page=1&per_page=25`. The maximum page size is 100.
`POST /api/v1/employees`	Send an `employee` object with business ID, name, department, country, designation, status, base_salary, currency and effective_date. Valid creation returns 201 and the employee's numeric ID.
`PATCH /api/v1/employees/42/salary` PUT is also routed.	Body: `{"salary":{"base_salary":"85000.25","currency":"USD","effective_date":"2026-10-13","confirmed":true}}`. Missing confirmation or invalid salary values are rejected.
`GET /api/v1/compensation_insights` `country, department, status`	`/api/v1/compensation_insights?department=Engineering&status=active`. Filters apply to every returned statistic. Salaries remain separated by currency.
`GET /api/v1/salary_analysis` `report, country, department, status, currency, page, per_page`	`report=roles` (default): role/country/currency group statistics. `report=outliers`: salaries above the configured quartile/IQR threshold. `report=below_department_average`: employees below their department/currency average.
Salary-analysis example	`/api/v1/salary_analysis?report=outliers&currency=USD&country=United%20States&page=1&per_page=8`. Cohorts use the full filtered population before pagination, not only the current page.

Responses and Errors

Read requests generally return `200`; login and employee creation return `201`; logout returns `204`. Missing/invalid authentication returns `401`. An authenticated user without the required role is rejected with `403`. Unknown employee IDs return `404`. Invalid values generally return `422`; malformed required request objects can return `400`. Login rate limits return `429`.

Frontend and Status Assets

The React portal is served at its own `/` URL, with sign-in protecting employee workflows. Its local origin is `http://127.0.0.1:5173`; the configured production origin is `https://employee-portal-frontend-1hdp.onrender.com`. The backend's default production host is `https://employeeportal-z31o.onrender.com`. Live availability and deployment of the newest code have not been verified.

Public status assets: `/api-status-theme.js` applies and saves the theme; `/api-mark.png` displays the brand; `/api-theme-sun.svg` and `/api-theme-moon.svg` provide toggle icons. They contain no employee data.

This reference covers application routes configured in `Backend/config/routes.rb`. Rails-generated Action Mailbox and Active Storage engine routes are framework infrastructure, not implemented employee features, and are excluded. No CSV/Excel export, simulation or review-schedule endpoint exists yet.

Salary Data-Quality API

Every route below requires the existing HR bearer session. Responses are marked `no-store`. Checks and decisions never change employee salaries.

Method	Route	Use
`POST`	`/api/v1/salary_data_quality_scans`	Run all eight checks across all employees. Returns 201 with checked employees, total/new findings, affected employees, cleared findings and start/completion times.
`GET`	`/api/v1/salary_data_anomalies`	Read a filtered, paginated finding queue and latest completed scan metadata. Active findings only by default; the UI initially selects Pending.
`GET`	`/api/v1/salary_data_anomalies/:id`	Read observed values, reference thresholds, detection snapshot, latest HR decision and a paginated immutable review history.
`PATCH` `PUT`	`/api/v1/salary_data_anomalies/:id`	Append an HR decision with notes and author/time. Returns 409 if the source changed or the finding cleared; rerun checks before reviewing.

Queue Filters and Pagination

`country, department, status, currency, q, review_status, include_cleared, page, per_page`. Employee status is active/inactive; review status is pending/investigating/confirmed_error/legitimate. `include_cleared` accepts true/false and defaults false. Default page size is 25, capped at 100. Detail pagination applies to review events, not employees. Filtered counts count findings; scan counters cover the entire database.

Decision Body and Errors

```
{"review":{"status":"investigating","notes":"Verifying the source record"}}
```

Confirmed error and Legitimate require nonblank notes; maximum 2,000 characters. Invalid values return 422, malformed review objects 400, unknown finding IDs 404 and stale source records 409. Unauthenticated requests return 401.

Rechecks and Deployment

Identical findings retain decisions and do not duplicate. Changed evidence creates a new Pending finding; old evidence and decisions stay available under Include cleared. Confirming an error does not clear it automatically: correct the salary and rescan. There is no scheduled scan. Apply migration `20261013000004` before using these routes; no seed or new environment setting is needed.

> **Scope:** suspicious values are review indicators, not proven mistakes or salary-fairness conclusions. Country/role contextual fairness remains deferred. Existing validation still blocks negative salaries and unsupported currencies.